NANYANG TECHNOLOGICAL UNIVERSITY

SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2 MH1301 Discrete Mathematics Tutorial 11

For the tutorial in Week 12 on 11 April, let us discuss the following questions.

Questions 1–3. Assume that $n \geq 7$,

1. How many (non-isomorphic) trees with n vertices are there in which one vertex has degree $n-1$?
2. How many (non-isomorphic) trees with n vertices are there in which one vertex has degree $n-2$?
3. How many (non-isomorphic) trees with n vertices are there in which one vertex has degree $n-3$?

Question 4. (Ex 11.1.8). Prove that a simple graph is a tree if and only if it is connected, but the deletion of any of its edges produces a graph that is not connected.

Question 5. (Ex 11.1.10). Which complete bipartite graphs $K_{m,n}$, where m and n are positive integers, are trees?

Question 6. (Ex. 11.1.11, 11.1.13).

(a) How many edges does a tree with 10000 vertices have?
(b) How many edges does a full binary tree with 10000 internal vertices have?

Question 7. (Ex 11.1.23). How many edges are there in a forest of t trees containing a total of n vertices?

Question 8. (Ex 11.1.19, 11.1.20). A **complete binary tree** is a full binary tree in which every leaf is at the same level.

(a) Construct a complete binary tree of height 4.
(b) How many vertices and how many leaves does a complete binary tree of height h have?

Answer.

1) 1.
2) 1.
3) 3.
5) $m = 1$ or $n = 1$.
6) 9999; 20000.
7) $n - t$.
8) 2^h.